Exercise: Atmosphere and Airspeeds

The indicated airspeed of the aircraft is $V_{IAS} = 320{,}8\ kts$. The atmosphere is assumed to be the standard atmosphere. The density of the surrounding air is $\tilde{\rho} = 0{,}916\,\frac{kg}{m^3}$.

For this type of aircraft, the following relation holds: $V_{CAS} = 1{,}01 \cdot (V_{IAS} - 2\ kts)$.

a) What is the geometric height of the aircraft?
b) Calculate the calibrated airspeed V_{CAS}.
c) Can we assume the air to be incompressible?
d) Calculate the difference between total and static pressure.
e) Calculate the equivalent airspeed V_{EAS}.
f) Calculate the true airspeed V_{TAS}.

Please use the following values in your calculation:

MSL density according to ISA standard atmosphere:	$\rho_{s,MSL} = 1.225\,\frac{kg}{m^3}$
MSL pressure according to ISA standard atmosphere:	$p_{s,MSL} = 1.01325e5\ Pa$
MSL temperature according to ISA standard atmosphere:	$T_{s,MSL} = 288.15\ K$
Gravitational acceleration:	$g_0 = 9.80665\,\frac{m}{s^2}$
MSL speed of sound:	$a_{MSL} = 340\,\frac{m}{s}$
Specific gas constant:	$R = 287.05\,\frac{J}{kg\ K}$
Polytropic exponent:	$n = 1.235$
Isentropic exponent:	$\kappa = 1{,}4$
Radius of the earth:	$r_E = \ 6356766m$